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Toilet with canned sump

Abstract

A sump for a toilet includes a first end of the sump, defined by a first boundary, and a second end of the sump, defined by a second boundary. The sump includes a channel running from the first end of the sump to the second end of the sump defined by a sump surface. The first end of the sump is configured to receive liquid and the second end of the sump is configured to supply liquid. The sump has a sump can-cut diameter corresponding to a diameter of the channel at the first end of the sump, a sump side radius extending from a center of the channel to an inner surface of the channel, and a sump depth corresponding to a largest vertical distance within the channel. The sump may be drain cast using a single continuous pathway.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2411301	12/1945	Jacques	N/A	N/A
4316864	12/1981	Bramwell	N/A	N/A
8032956	12/2010	Prokopenko	4/425	E03D 11/08
2003/0115664	12/2002	Kosugi et al.	N/A	N/A
2019/0106872	12/2018	Grover et al.	N/A	N/A
2020/0102729	12/2019	Lambert et al.	N/A	N/A
2020/0332509	12/2019	Smith et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105365030	12/2015	CN	N/A
0572280	12/1992	EP	N/A
2221252	12/1973	FR	N/A
489438	12/1937	GB	N/A
H068475	12/1993	JP	N/A
2001303648	12/2000	JP	N/A
2002348940	12/2001	JP	N/A
2004011414	12/2003	JP	N/A
5174504	12/2012	JP	N/A
2015194004	12/2014	JP	N/A
102017460	12/2018	KR	N/A
102074681	12/2019	KR	N/A
WO-2012140607	12/2011	WO	B28B 1/002
2017056639	12/2016	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report from European Patent Application No. 22 186 645.2, dated Jun. 24, 2025, 7 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority benefit of Provisional Application No. 63/226,920 filed on Jul. 29, 2021, which is hereby incorporated by reference in its entirety.

FIELD

(1) The present application relates generally to plumbing fixtures. More specifically, the present disclosure relates to plumbing fixtures that have a drain-casted sump.

BACKGROUND

(2) Generally speaking, plumbing fixtures, such as toilets and urinals include a sump in which liquid and/or waste are collected. The sump being fluidly connected to a bowl of the toilet or urinal through which liquid and/or other waste may enter the sump. The sump is also fluidly connected to a trapway leading to a drain or sewer. Liquid and/or other waste flow from the sump through the trapway to a drain or sewer during a flushing operation. Generally, a jet orifice is provided in the sump. The jet orifice fluidly connects a drain channel, which supplies water to the jet orifice, to the sump. Water supplied from the drain channel, through the jet orifice, and to the sump facilitates the removal of liquid and/or other waste from the plumbing fixture during a flushing process.

(3) Plumbing fixtures including a sump are typically manufactured using a solid cast process. In the solid cast manufacturing process, the bowl is solid cast using a mold and then holes fluidly connecting the sump to the drain channel and bowl are manually punched. Manually punching the holes that connect the sump to the drain channel and bowl require that there is sufficient room in the sump for the punching device to be inserted and properly positioned to punch the holes, thereby limiting the location and orientation of the jet orifice and the shape of the sump. Plumbing fixtures manufactured using a solid cast process have substantially rectangular shaped sump with few, if any, smooth continuous curved surfaces. Additionally, manually punching holes connecting the sump to the bowl and jet orifice may lead to material yield and inconsistent shape and location of the punched holes, reducing the flushing efficiency of the plumbing fixture.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments are described herein with reference to the following drawings.

(2) FIG. 1 illustrates a toilet with a canned sump according to an exemplary embodiment of the present disclosure.

(3) FIG. 2 illustrates a right-side perspective view of a solid cast toilet.

(4) FIG. 3 illustrates a right-side perspective view of a toilet according to an exemplary embodiment of the present disclosure.

(5) FIG. 4 illustrates a front perspective view of the toilet of FIG. 3 according to an exemplary embodiment of the present disclosure.

(6) FIG. 5 illustrates a zoomed-in perspective view of the toilet of FIGS. 3 and 4 according to an exemplary embodiment of the present disclosure.

(7) FIG. 6 illustrates a right-side perspective view of a toilet according to an exemplary embodiment of the present disclosure.

(8) FIG. 7 illustrates a side view of a toilet including a drain cast sump with dimples according to an embodiment of the present disclosure.

- (9) FIG. 8 illustrates a front view of the toilet of FIG. 7 including a drain cast sump with dimples.
- (10) FIG. 9 illustrates a drain cast method of forming the sump according to an exemplary embodiment of the present disclosure.
- (11) FIG. 10 illustrates a can and a toilet according to an exemplary embodiment of the present disclosure.
- (12) FIG. 11 is a side view of a traditional solid cast toilet overlayed on top of a toilet including a drain cast sump according to an exemplary embodiment of the present disclosure.
- (13) FIG. 12 is a front view of the traditional solid cast toilet overlayed on top of a toilet including a drain cast sump of FIG. 11.

DETAILED DESCRIPTION

(14) Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details and methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

(15) Generally speaking, sump waste removal fixtures, such as a toilet or urinal, include a sump in which liquid and waste in the toilet or urinal collect. The sump waste removal fixture generally includes an orifice, which may be a jet orifice, for introducing water into the trapway of the fixture. The water may be used to assist in the draining of the contents of the bowl and sump of the plumbing fixture. The contents of the bowl and sump may be drained through the trapway to a drain pipe or sewer line. Some plumbing fixtures are made from a vitreous material by a solid casting process, where a slip material or tube is utilized to form a hollow trapway within the plumbing fixture when the plumbing fixture is cast. After the plumbing fixture is cast, a sump mouth and a jet orifice are typically formed through a solid casted wall of the plumbing fixture by manually using a punch to fluidly couple the sump to the trapway and drain pipe and fluidly couple the jet orifice to the sump. A fluid channel typically directs fluid from a fluid supply source to the jet orifice to assist with draining. However, a significant amount of clearance is required to form the sump mouth and jet orifice using a manual punch. Accordingly, the shape of the sump and location and orientation of the sump mouth and jet orifice are limited. In addition, forming the sump mouth and jet orifice with a punch can result in significant material yield, and can result in inconsistent sizes or shapes of the jet orifice, thereby negatively impacting drain performance.

(16) Referring generally to FIGS. 1 and 3-8, disclosed herein are plumbing fixtures (e.g., toilets, urinals, etc.) having a sump formed using a drain casting process instead of a manual punching operation. The drain casting process, advantageously, allows for the sump to be cast in many shapes not feasible using the solid cast method. The drain casting process allows the sump to be cast having smoother and more continuous walls, improving washing and waste removal. Additionally, the drain casting process, advantageously, allows the jet orifice to be located in the sump near the inlet of the trapway, improving fluid flow and draining of the trapway. Forming the sump mouth and the jet orifice using a drain-casting process instead of a punch provides a more consistent shape of the mouth and drain orifice with limited or no material yield during the manufacturing process. Additionally, the drain casting process may be used to improve manufacturability of plumbing fixtures having a sump.

(17) The following description focuses primarily on a drain-casted sump for a toilet. However, it should be appreciated that the disclosed sump structure and drain casting process can be applied to a variety of other types of plumbing fixtures and trapways, such as a urinal, lavatories, or other plumbing fixtures that include a sump, trapway, and a drain.

(18) Referring to FIG. 1, a toilet 1100 with a canned sump is illustrated according to an exemplary embodiment of the present disclosure. The toilet 1100 may include a tank (e.g., container, reservoir, etc.), shown as a tank 1102, and a pedestal (e.g., base, stand, support, etc.), shown as a pedestal 1104. The tank 1102 may be coupled to, and supported by, the pedestal 1104, which may be

positioned on a floor. In some embodiments, the tank **1102** and the pedestal **1104** may be formed together as a single component. The tank **1102** is configured to receive water (e.g., via a fill valve of the toilet **1100**, etc.) and store the water in between flushes. The pedestal **1104** includes a bowl **1105** and may be configured to receive the water from the tank **1102** to flush contents of bowl into a sewage line. In some embodiments, the pedestal **1104** may be mounted on the wall of a lavatory and the bowl **1105** may be configured to receive water from a fluid supply source such as a household water supply.

(19) The bowl **1105** of the pedestal **1104** includes a sump (e.g., a receptacle) and an outlet opening, wherein water and waste is collected in the sump until being removed through the outlet opening, such as when the contents of the bowl **1105** are flushed into a sewage line. The toilet **1100** further includes a trapway, the trapway being fluidly connected to the bowl **1105** via the sump. The trapway fluidly connects the sump to the outlet opening.

(20) Referring to FIG. 2, a toilet **100** including a sump **115** formed using a solid casting method is shown. The toilet **100** includes a bowl **110** having an inlet **112** defined by a bowl surface **111**. The toilet **100** further includes a sump **115** defined by a sump surface **116** extending from an upper boundary **117** (i.e., a sump mouth) located at the bowl surface **111** to a lower boundary **118** located at a trapway **120**. The trapway **120** is defined by a trapway surface **121**. The sump **115** fluidly connects the bowl **110** and the trapway **120**. The sump surface **116** extends from the bowl surface **111** to the trapway surface **121** and is coextensive with the bowl surface **111** and trapway surface **121**. The toilet **100** further includes a drain channel **130** extending partially in front of the sump **115**. In some embodiments the toilet **100** may have two drain channels connecting partially in front of the sump **115**. In some embodiments, the drain channel **130** may extend partially below the sump **115**. The drain channel **130** may be fluidly coupled to a fluid supply source, such as a household water supply. The drain channel **130** includes a drain opening that is defined by a drain surface. The drain channel **130** terminates at the sump surface **116** to define a jet orifice **133** (e.g., sump jet orifice, etc.). The jet orifice **133** may be located on a front or a bottom surface of the sump **115**. The toilet **100** is configured such that the drain channel **130** can direct a flow of water from the water supply source to the jet orifice **133**, where the flow of water can be introduced to the sump **115** to, for example, to help prime a siphon in the trapway **120** to induce a flush cycle and empty the contents of the bowl **110**.

(21) In the toilet **100**, according to the casting method of FIG. 2, a significant amount of clearance is required to manually punch the holes for the sump mouth, located at the upper boundary **117** of the sump **115** and the jet orifice **133**. Accordingly, the geometry of the sump **115** is limited to that which will allow the punch to fit inside the sump **115** to manually punch the sump mouth and jet orifice **133**. Similarly, the location and orientation of the sump mouth and jet orifice **133** are limited to those locations and orientations that can be manually punched. Additionally, manual punching can lead to significant material yield and inconsistent size and locations of the jet orifice, negatively impacting drain performance.

(22) Referring to FIGS. 3-5 a toilet **300** according to an exemplary embodiment of the present disclosure is shown. The toilet **300** includes a bowl **310** having an inlet **312** defined by a bowl surface **311**. The toilet further includes a sump **315** defined by a sump surface **316** that extends from an interior surface of the bowl **310** at an upper boundary **317** (e.g., sump mouth) to an interior surface of the trapway **320** at a lower boundary **318**. The sump surface **316** may be an inside surface of the channel within the sump **315**. The sump fluidly connects the bowl **310** and the trapway **320**. The sump surface **316** extends from, and is coextensive with, the bowl surface **311**. The sump **315** may have a sump depth **420** corresponding to a largest vertical distance within the channel of the sump **315** fluidly connecting the bowl **310** and the trapway **320**. The sump **315** may have a sump can cut diameter **410** corresponding to a diameter of the channel formed within the sump **315** at a first end of the sump (e.g., a sump mouth) formed by a boundary between the bowl **310** and the sump **315**. The boundary between the bowl **310** and the sump **316** may be a location at

which a can is used to remove a clay cap, fluidly connecting the bowl **310** and the sump **315**. The sump can cut diameter **410** is the diameter (e.g., distance) of the sump surface **316** at the point which the bowl **310** and the sump **315** meet. Referring to FIG. **4**, the sump **315** may have a sump side radius **430** extending from a center of the channel within the sump **315** to an inside surface of the channel (e.g., sump surface **316**).

(23) The sump surface **316** is coextensive with a trapway surface **322** that defines a trapway opening. The trapway opening extends to an outlet port **321** for connecting the toilet **300** to a sewer or drain. In this exemplary embodiment, the toilet **300** including trapway **320** is configured as a siphonic toilet, whereby the contents of the bowl **310** are emptied by creating a siphon effect in the trapway **320**, the details of which are discussed below. According to another exemplary embodiment, the toilet **300** may be configured as a blowout style jetted toilet. The toilet **300** further includes a drain channel **330** extending partially in front of the sump **315**. In some embodiments the toilet **300** may include a left drain channel **331** and a right drain channel **332**. In other embodiments, the toilet **300** may only include one drain channel **330**. In some embodiments, the drain channel **330** may extend partially below the sump **315**. The drain channel **330** may be fluidly coupled to a fluid supply source, such as a household water supply. The drain channel **330** includes a drain opening that is defined by a drain surface. The drain channel terminates at the sump surface **316** to define a jet orifice **333** (e.g., sump jet orifice, etc.). The jet orifice **333** may be located on a front or bottom surface of the sump **315**. The toilet **300** is configured such that the drain channel **330** can direct a flow of water from a fluid supply source to the jet orifice **333**, where the flow of the water can be introduced to the sump **315** to, for example, help to prime a siphon in the trapway **320** to induce a flush cycle and empty the contents of the bowl **310**. In some embodiments, the toilet **300** may include two or more jet orifices. The two or more jet orifices may both be included in a continuous drain cast channel used to form the toilet **300**.

(24) The toilet **300** according to FIGS. **3-5** may be cast using a drain casting method, such as the method as shown in FIG. **8**. Drain casting the toilet **300** advantageously allows the sump **315** to be cast in a wide variety of shapes as a punch does not need to be inserted into the sump to manually punch holes connecting the sump to the jet orifice and/or the toilet bowl. Additionally, drain casting the sump allows the bowl, drain channel, and sump all to be fluidly connected by a single operation of opening the sump mouth using a can. The toilet **300** may be advantageously cast to have a specific ratio of can-cut diameter **410** over sump depth **420** and/or cast to have a specific ratio of can-cut diameter **410** over sump side radius **430**. The flushing efficiency of the toilet **300** may be improved by having a sump **315** having a specific ratio of can-cut diameter **410** over sump depth **420** and/or a specific ratio of can-cut diameter **410** over sump side radius **430**. The flushing efficiency of the toilet **300** may be measured using series of tests and/or models, for example adding granules/particles to the contents of a toilet bowl and determining how many granules/particles remain in the bowl after a flushing cycle, measuring the flow rate of liquid through the sump and/or trapway during a flushing cycle, and the like.

(25) In some examples, the ratio of the sump can-cut diameter over the sump radius may be 2.2 to 2.4. In other examples, the ratio of sump can-cut diameter over the sump side radius may be 2.4 to 2.6. In other examples, the ratio of sump can-cut diameter over the sump side radius may be greater than 2.6. In some examples, the ratio of sump can-cut diameter over the sump depth is 1 to 1.2. In other examples, the ratio of sump can-cut diameter over the sump depth is 1.2 to 1.4. In some examples, the ratio of sump can-cut diameter over the sump depth is greater than 1.4. In some examples, the ratio of sump can-cut diameter over sump depth is greater than 1.15 and the ration of the sump can-cut diameter over sump side radius is greater than 2.3.

(26) Referring to FIG. **6**, a toilet **200** including a drain cast sump according to an exemplary embodiment of the present disclosure is shown. The toilet **200** includes a bowl **210** having an inlet **212** defined by a bowl surface **211**. The toilet further includes a sump **215** defined by a sump surface **216** that extends from the bowl **210** at an upper boundary **217** to a trapway **220** and a lower

boundary **218**. The sump **215** fluidly connects the bowl **210** and the trapway **220**. The sump surface **216** extends from, and is coextensive with, the bowl surface **211**. The sump **215** has a sump depth **420** and a sump can-cut diameter **410**. The sump can-cut diameter **410** is the diameter of the sump surface **316** at the point at which the sump **215** and the bowl **210** meet. The sump **215** further has a sump side radius extending from a center of the sump **215** to an inside surface of the sump (not shown).

(27) The sump surface **216** is coextensive with the trapway surface **322** that defines a trapway opening. The trapway extends from the sump **215** to an outlet **221** for connecting the toilet **200** to a sewer or drain. In this embodiment, the toilet **200** has one drain channel **230**, extending partially in front of the sump **215**. In this embodiment, the drain channel **230** fluidly connects a fluid supply source to a jet orifice **233** located at the sump surface **216**. The drain channel **230** is configured to direct a flow of liquid from the water supply source to the sump **215**.

(28) Referring to FIGS. **7** and **8**, a toilet **500** including a drain cast sump with dimples is shown according to another exemplary embodiment of the present disclosure. The toilet **500** includes a bowl **510** having an inlet **512** defined by a bowl surface **511**. The toilet **500** further includes a sump **515** defined by a sump wall **516** that extends from the bowl **510** at an upper boundary **517** to a trapway **520** at a lower boundary **518**. The sump **515** fluidly connects the bowl **510** and the trapway **520**. The sump surface **516** extends from, and is coextensive with, the bowl surface **511**. The sump **515** has a sump depth **420** and a sump can-cut diameter **410**. The sump can-cut diameter **410** is the diameter of the sump surface **516** at the point at which the sump **515** and the bowl **510** meet. Referring to FIG. **7B**, the sump **515** has a sump side radius **430** extending as a perpendicular distance (perpendicular from the direction of flow) from the center of the sump **515** to the inside surface or outside surface of the sump **515**. The sump side radius **430** may be measured from one or more dimples **570** as described below.

(29) The sump surface **516** is coextensive with the trapway surface **522** that defines a trapway opening. The sump **515** and the trapway **520** may have a substantially circular cross section. The trapway **520** extends from the sump **515** to an outlet **521** for connecting the toilet **500** to a sewer or a drain. The toilet **500** includes a drain channel **530**. In some embodiments, the toilet **500** may include two drain channels. In some embodiments, the drain channel **530** may extend partially in front of the sump **515**. In some embodiments, the drain channel **530** may extend partially below the sump **515**. The drain channel **530** may be fluidly coupled to a fluid supply source and includes a drain opening that is defined by a drain surface. In some embodiments, the toilet **500** may have a left drain channel **531** and a right drain channel **532**. The drain channel **530** may connect the fluid supply source to a jet orifice **533** located at the sump surface **516**. The jet orifice **533** may be located on a front of the sump surface **516**. The jet orifice **533** may be located on the bottom of the sump surface **516**. The drain channel **530** is configured to direct a flow of water from the fluid supply source to the sump **515** to, for example, induce a flush cycle.

(30) The toilet **500**, further includes one or more dimples **570** located on the sump **515** and/or trapway **520**. The dimples **570** may be substantially flat portions of the sump **515** and/or trapway **520** in which the sump surface **516** and/or the trapway surface **522** are located closer to the center of a sump **515** and/or trapway **520** having an otherwise circular cross section. The dimples **570** may be located completely on the sump **515**, completely on the trapway **520**, and/or on both the sump **515** and the trapway **520**. Referring to the perspective view of FIG. **8**, in some embodiments, the sump **515** and/or trapway **520** may have two dimples **570a**, **570b**, one located on each of the left and right side of the sump. The dimples **570a**, **570b** may improve the flushing efficiency of the toilet **500**. The dimples **570a**, **570b** may cause more liquid entering the sump **515** through the drain channel **530** to contact the sump surface **516** and/or the trapway surface **522** improving surface washing and waste removal during a flushing cycle.

(31) Referring to FIG. **9**, a flow chart for a drain-cast method of forming a sump according to an exemplary embodiment of the present disclosure is provided. The method of FIG. **9** may be used to

form any of the sumps **315**, **215**, **515** described herein. The method of FIG. **9** is described below with reference to the sump **315** of FIGS. **3-5**. Additional, different, or fewer acts may be provided. (32) At act **S101**, a mold having the basic shape and structure of the toilet **300** is filled with liquid clay slip. The mold is oriented such that the front portion **360** is located on the bottom of the mold with the rear portion **350** oriented in an upward direction above the front portion **360**. During the initial forming step **S101**, the toilet **300** includes a circular clay cap **370** disposed in the trapway **320** at the boundary between the co-extensive walls of the bowl surface **311** and the sump **315**. The circular clay cap **370** blocks the flow of liquid slip between the toilet bowl surface **311** and the sump **315**. During the initial forming step **S101**, the drain channel **330**, including the mouth **333** of the drain channel are open and fluidly connected to the sump **315** and the trapway **320** so as to define a drain cast pathway of the toilet **300**. The liquid clay slip extends through the drain channel **330**, the jet orifice **333** of the drain channel, the sump **315**, the trapway **320**, to an outlet port **321**. In other words, the toilet **300** is designed such that the drain channel **330**, jet orifice **333** of the drain channel, the sump **315**, and the trapway **320** cooperatively define a continuous drain cast pathway of the toilet **300**.

(33) At act **S103**, the liquid slip may set up in the cast to form the various solid cast walls of the toilet **300**. The mold may draw moisture from the liquid clay slip. As moisture is drawn from the liquid clay slip, a thickness forming the solid cast walls of the toilet **300** may accumulate within the mold. The liquid clay slip may set for a predetermined period of time corresponding to a desired thickness of the various solid cast walls of the toilet **300**.

(34) At act **S105**, the mold pins are removed such that the liquid slip remaining in the mold drains from the drain channel **330**, jet orifice **333** of the drain channel, sump **315** and trapway **320** of the mold. During the third step act **S105** additional forming parts (e.g., funnels for directing liquid slip into the molds, etc.) may be removed from the mold. Further, in some embodiments, during act **S105**, the mold may be rotated (e.g., an orientation of the mold may be changed). In some examples, the mold may be rotated to assist in draining the liquid clay slip from the drain channel **33**, jet orifice **333** of the drain channel, sump **315**, and the trapway **320**. In other embodiments, the mold may be rotated in preparation for act **S107**.

(35) At act **S107**, a can may be used to cut out the circular clay cap **370** disposed between the bowl **310** and the sump **315**. FIG. **10** described below illustrates toilet and a can according to an exemplary embodiment of the present disclosure. The can may be inserted into the bowl **310** of the toilet and positioned such that an opening of the can is aligned with the clay cap **370**. The can may be pushed into the clay cap **370**, cutting the clay cap **370** from between the bowl **310** and the sump **315**. The can may then be withdrawn from between the bowl **310** and the sump **315**. The clay cap **370** may be removed as the can is withdrawn from between the bowl **310** and the sump **315**.

(36) At act **S107**, removing the clay cap **370** opens the sump mouth fluidly connecting the bowl **310** and the sump **315**. In some embodiments, the can may be rotated or twisted as it is inserted into the clay cap **370** between the bowl **310** and the sump **315**. In some embodiments, the can may be rotated or twisted as it is withdrawn from between the bowl **310** and the sump **315**. The can may be rotated during insertion and/or withdrawal to improve the quality of the cut as the clay cap **370** is removed. Improving the quality of the cut may improve the surface quality of the connection (e.g., sump mouth, can cut diameter) between the bowl **310** and the sump **315**. Thus, the quality of a fluid flowing through the connection (e.g., sump mouth, can cut diameter) may be improved.

(37) In some embodiments, a caster may manually cut out the clay cap **370** between the bowl **310** and the sump **315**. In some examples, a tool having the shape of the bowl **310** may be used to position the can at the correct location in the bowl to cut out the circular clay cap **370**. In other embodiments, an automated machine or device may use the can to cut out the circular clay cap. During act **S107**, additional forming parts remaining in the mold (e.g., a center part, a core, a front part, etc.) may be removed.

(38) At act **S109**, the toilet **300** is dried for a period of time. The toilet **300** may be dried for a

predetermined period of time allowing moisture to be wicked from the toilet **300**. At act **S111**, the toilet may be sprayed with a glaze. At act **S113**, the toilet may be fired in a kiln to form the final toilet **300**. Additional, fewer, and/or different steps may be included.

(39) FIG. **10** illustrates a toilet **300** and a can **700** used to open the sump mouth according to an exemplary embodiment of the present disclosure. FIG. **10** illustrates the can **700** and toilet **300** after the can **700** has been used to remove the circular clay cap disposed between the bowl **310** and the sump **315** of the toilet **300**. FIG. **10** may illustrate the can **700** being withdrawn from the bowl **310** at act **S107** of the method of FIG. **9** described above. In some embodiments, as illustrated in FIG. **10**, a caster may manually open the sump mouth using the can **700**. In some embodiments, a tool having a shape matching a portion of the bowl may be used to position the can **700** in the correct position before opening the sump mouth may be used.

(40) Referring to FIGS. **11** and **12**, a side view and a front view, respectively, of a solid cast toilet **600** overlaid on a drain cast toilet **610**, according to an exemplary embodiment of the present disclosure, is shown. FIG. **11** illustrates a right-side perspective view of a solid cast toilet **600** overlaid on a drain cast toilet **610** according to an exemplary embodiment of the present disclosure. FIGS. **11** and **12** illustrate differences in geometry of the solid cast toilet **600** and the drain cast toilet **610**. During manufacture of the drain cast toilet **610**, opening the sump mouth (e.g., cutting out/removing the circular clay cap) using a can, as opposed to a punch, and not having to open the jet orifice allows the bowl and sump to be manufactured having with a wider variety of shapes. In the drain cast toilet **610**, the bowl and sump may be manufactured with a wider variety of shapes because a punch does not need to be inserted into the sump and only one operation is performed (i.e., opening the sump mouth with the can) as opposed to two operations in the solid cast toilet (i.e., punching the connection between the sump and the drain channel and punching the connecting between the sump and the bowl). For example, the drain cast toilet **610** as disclosed herein may have more vertical surfaces, improving surface washing and waste removal.

(41) Plumbing fixtures employing the apparatuses and methods as disclosed above may additionally be manufactured using a toilet rim, for example a tile-on rim, that is cast separately from other components of the plumbing fixture. The tile-on rim may be a solid annular member disposed around a top of the toilet bowl, surrounding an opening in the bowl. The tile-on rim may be planar including a flat top and/or bottom surface. The tile-on rim may be coupled to the bowl by way of a bonding material (e.g., adhesives, cements, and the like) and/or one or more fasteners. For example, the bonding material may be clay, ceramic, or vitreous (e.g., a “slip-roll” clay) that is applied to one or both of the tile-on rim and the bowl as a paste or malleable solid, such as a semi-liquid and a semi-solid material, then heated (e.g., in a kiln or oven) to permanently fuse the tile-on rim and the bowl together.

(42) The tile-on rim may be disposed entirely above a water channel formed in the bowl (e.g., a water channel configured to supply water to the bowl). The tile-on rim may be formable by a molding or casting process. For example, the tile-on rim may be made of a porcelain or vitreous material. The tile-on rim may include a back portion extending from a rear of the annular portion. The back portion may be configured to receive fasteners for securing the tile-on rim to the toilet and/or fasteners for securing a tank to the rim. The annular portion of the rim may be configured to receive a seat for supporting a user of the toilet, the seat may be disposed above the rim.

(43) The tile-on rim couples to and abuts an upper surface of the bowl. In some embodiments, the tile-on rim may be affixed to the bowl using bonding material and/or one or more fasteners. The bonding material may be applied around a portion of or the entire perimeter of the surface between the tile-on rim and the structure. In some examples, the tile-on rim may define a top of a water channel formed in the bowl (e.g., a water channel configured to supply water to the bowl). The use of tile-on rims with embodiments of the present disclosure may further improve manufacturability of the plumbing fixtures. For example, the toilets may be manufactured more quickly, or the number of steps that must be performed manually may be reduced.

(44) As utilized herein, the terms “approximately,” “about,” “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(45) It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(46) The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

(47) The term “or,” as used herein, is used in its inclusive sense (and not in its exclusive sense) so that when used to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is understood to convey that an element may be either X, Y, Z; X and Y; X and Z; Y and Z; or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

(48) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(49) Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(50) It is important to note that the construction and arrangement of the system as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other

embodiments disclosed herein.

(51) When a component, element, device, or the like of the present disclosure is described as having a purpose or performing an operation, function, or the like, the component, device, or element should be considered herein as being “configured to” meet that purpose or to perform that operation or function.

Claims

1. An assembly for a toilet, including a sump, the sump comprising: a first end of the sump, defined by a first boundary; a second end of the sump, defined by a second boundary; a channel running from the first end of the sump to the second end of the sump defined by a sump surface; a drain channel; and a jet orifice fluidly connected to the drain channel and the sump, the jet orifice configured to deliver liquid to the sump, wherein the first end of the sump is configured to receive liquid and the second end of the sump is configured to supply liquid, wherein the sump has a sump can-cut diameter corresponding to a diameter of the channel at the first end of the sump, a sump side radius extending from a center of the channel to an inside surface of the channel, and a sump depth corresponding to a largest vertical distance within the channel, and wherein the drain channel, jet orifice, and sump are drain cast using a single continuous drain cast pathway fluidly connecting the drain channel, the jet orifice, and the sump, such that a smooth and continuous curved surface is formed by the drain channel, the jet orifice, and the sump.
2. The assembly of claim 1, wherein the ratio of the sump can-cut diameter over the sump side radius is 2.2 to 2.4.
3. The assembly of claim 1, wherein the ratio of the sump can-cut diameter over the sump side radius is 2.4 to 2.6.
4. The assembly of claim 1, wherein the ratio of the sump can-cut diameter over the sump side radius is greater than 2.6.
5. The assembly of claim 1, wherein the ratio of the sump can-cut diameter over the sump depth is 1.0 to 1.2.
6. The assembly of claim 1, wherein the ratio of the sump can-cut diameter over the sump depth is 1.2 to 1.4.
7. The assembly of claim 1, wherein the ratio of the sump can-cut diameter over the sump depth is greater than 1.4.
8. The assembly of claim 1, wherein the ratio of the sump can-cut diameter over the sump depth is greater than 1.15 and the ratio of sump can-cut diameter over sump side radius is greater than 2.3.
9. The assembly of claim 1, further comprising: at least one dimple located on the sump, wherein the at least one dimple reduces the sump side radius along a portion of the sump.
10. A toilet, comprising: a bowl; a sump including a channel fluidly connected to the bowl, the sump having a can-cut diameter corresponding to a diameter of the channel at a location where the channel is fluidly connected to the bowl, a sump side radius extending from a center of the channel to an inside surface of the channel, and a sump depth corresponding to a largest vertical distance within the channel; a drain channel; a jet orifice fluidly connected to the drain channel and the sump, the jet orifice configured to deliver the liquid to the sump; and a trapway, fluidly connected to the sump, wherein, the drain channel, jet orifice, sump, and trapway are drain cast using a single continuous drain cast pathway fluidly connecting the drain channel, the jet orifice, the sump, and the trapway, such that a smooth and continuous curved surface is formed by the drain channel, jet orifice, and sump.
11. The toilet of claim 10, wherein the ratio of the sump can-cut diameter over the sump side radius is 2.4 to 2.6.
12. The toilet of claim 10, wherein the ratio of the sump can-cut diameter over the sump side radius is greater than 2.6.

13. The toilet of claim 10, wherein the ratio of the sump can-cut diameter over the sump depth is 1.2 to 1.4.
 14. The toilet of claim 10, wherein the ratio of the sump can-cut diameter over the sump depth is greater than 1.4.
 15. The toilet of claim 10, wherein the ratio of the sump can-cut diameter over the sump depth is greater than 1.15 and the ratio of sump can-cut diameter over sump side radius is greater than 2.3.
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